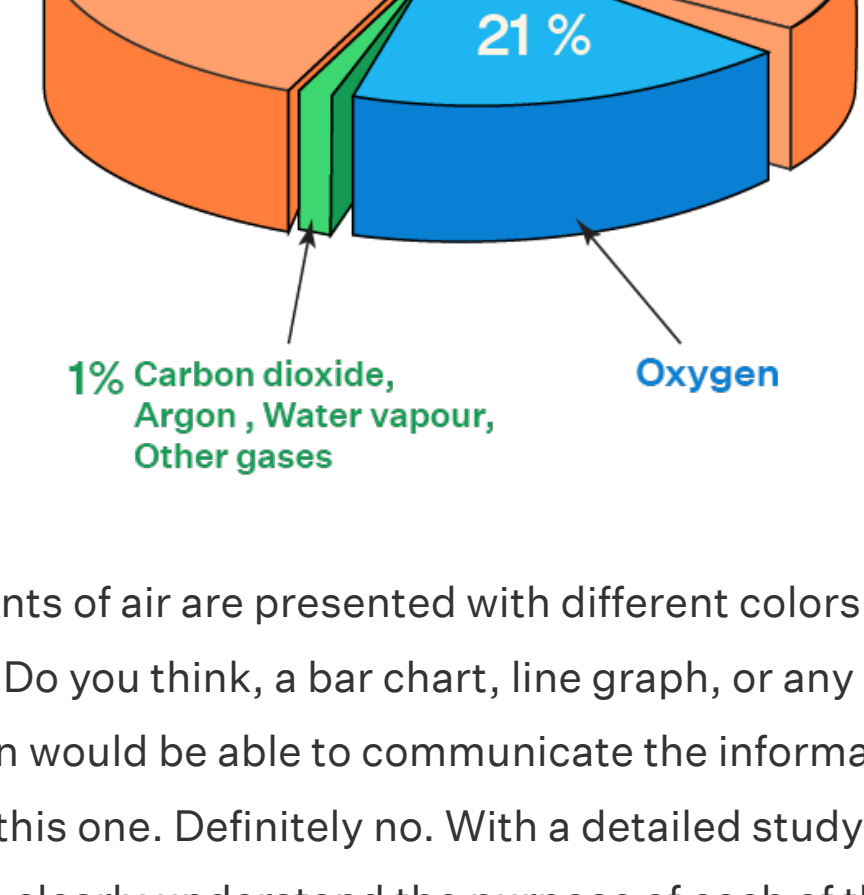


Data Handling

Data handling is considered one of the most important topics in statistics as it deals with collecting sets of data, maintaining security, and the preservation of the research data. The data here is a set of numbers that help in analyzing that particular set or sets of data. Data handling can be represented visually in the form of graphs. Let us learn more about this interesting concept, the different graphs used, and solve a few examples for better understanding.

Definition of Data Handling

Data Handling is the process of gathering, recording, and presenting information in a way that is helpful to analyze, make predictions and choices. Anything that can be grouped based on certain comparable parameters can be thought of as [data](#). Parameters mean the context in which the comparison is made between the objects. Data handling usually represent in the form of pictographs, bar graphs, pie charts, [histograms](#), line graphs, [stem and leaf plots](#), etc. All of them have a different purpose to serve. Have a look at the composition of the air that we have learned about in our science classes.



The constituents of air are presented with different colors in the form of parts of a pie. Do you think, a bar chart, line graph, or any other graphical representation would be able to communicate the information as effectively as this one. Definitely no. With a detailed study of each of them, you can clearly understand the purpose of each of them and use them suitably.

Types of Data

Data handling is performed depending on the types of data. Data is classified into two types, such as Quantitative Data and Qualitative Data. Quantitative data gives numerical information, while qualitative data gives descriptive information about anything. Quantitative can be either discrete or continuous data.

Important Terms in Data Handling

In data handling, there are 4 important terms or most frequently used terms that make it simple to understand the concept better. The terms are:

- **Data:** It is the collection of numerical figures of any kind of information
- **Raw Data:** The observation gathered initially is called the raw data.
- **Range:** It is the difference between the highest and lowest values in the data collection.
- **Statistics:** It deals with the collection, representation, analysis, and interpretation of numerical data.

Steps Involved in Data Handling

Following are the steps to follow in data handling:

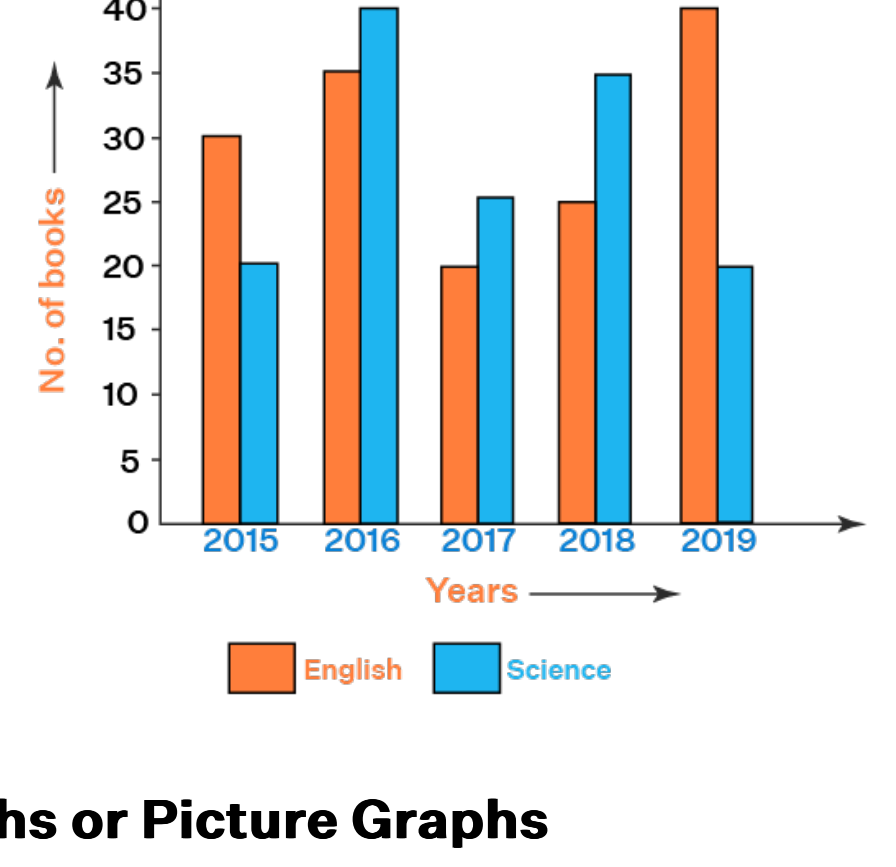
Steps	Details
Purpose	The problem or purpose is identified and well defined
Collection of Data	Data relevant to the purpose is collected.
Presentation of Data	The collected data is to be presented in a form that is meaningful and easy to understand. It could be in the form of a simple table or tally marks etc.
Graphical Representation of Data	Visual representation makes analysis and understanding of trends quicker and has a much greater impact.
Analyzing the Data	The data undergoes inspection to derive useful and necessary information that helps in taking further actions.
Conclusion/Inference	Here we provide a solution to our problem statement based on the analysis of the data.

Graphical Representation of Data Handling

Data handling can be represented in a number of graphical ways. Here is a list of various types of graphical representations of data that are very effective in data handling.

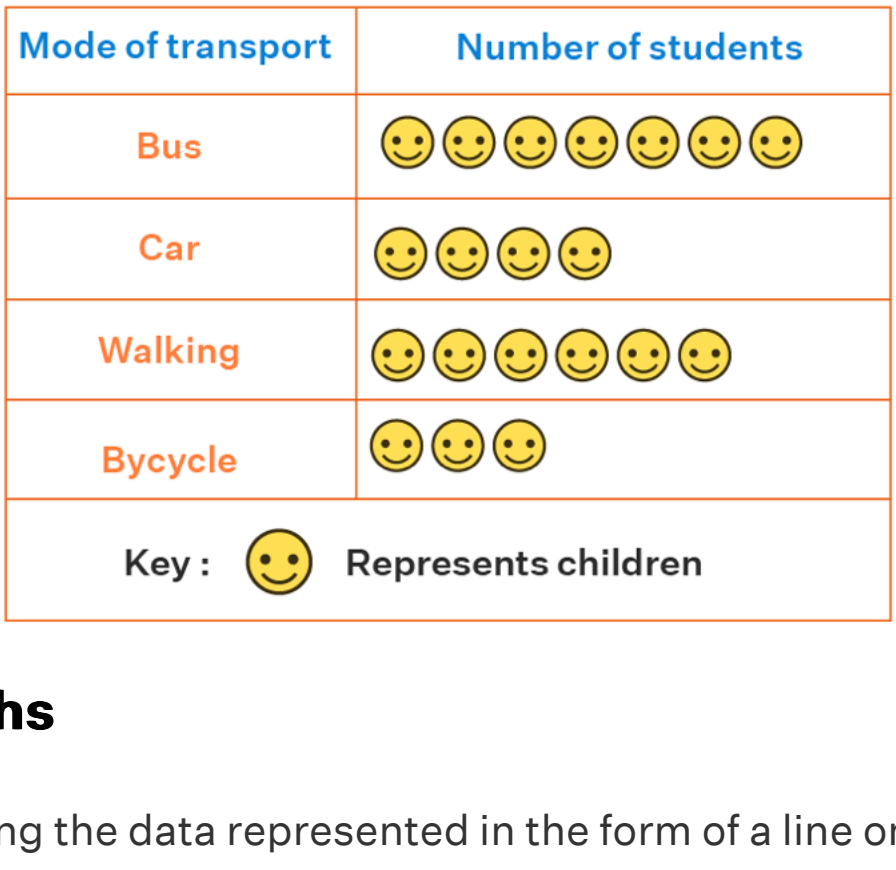
Bar Graphs

[Bar graphs](#) represent data in the form of vertical or horizontal bars showing data with rectangular bars and the heights of bars are proportional to the values that they represent. Bar graphs help in the comparison of data and this type of graph is most widely used in statistics. Look at the image below as an example.



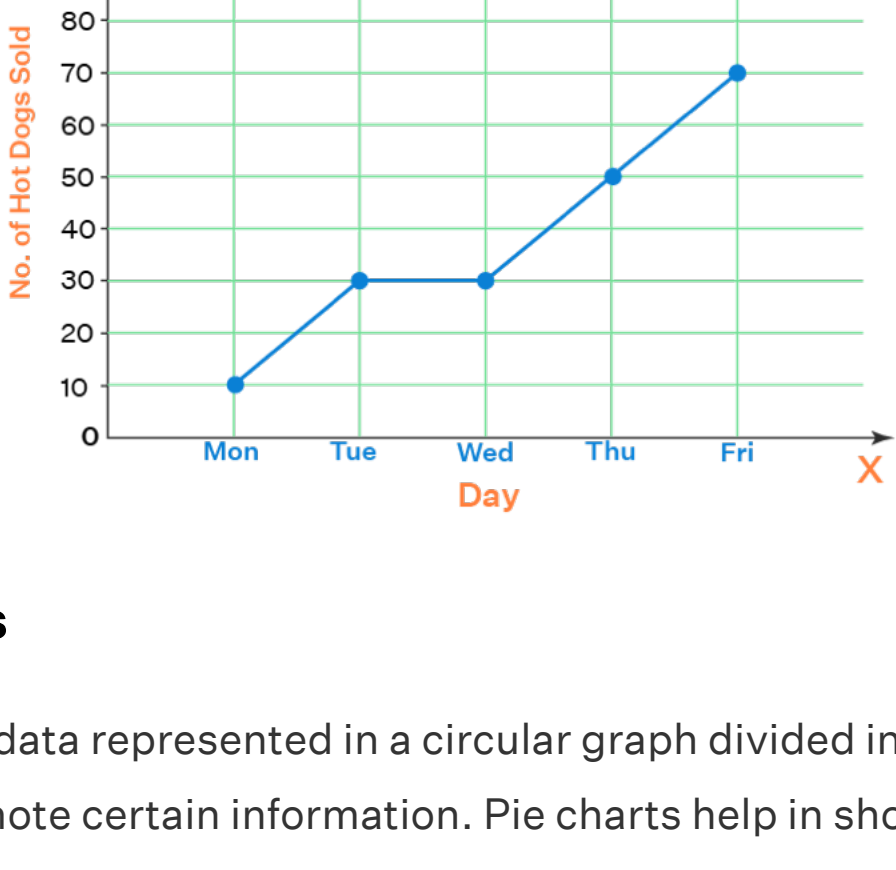
Pictographs or Picture Graphs

[Pictograph](#) is a type of graph where information is represented in the form of pictures, icons, or symbols. It is the simplest form of representing data in statistics and data handling. Since the use of images and symbols are more in a pictograph, interpreting data is made easy along with representing a large number of data. Look at the example below for a better understanding.



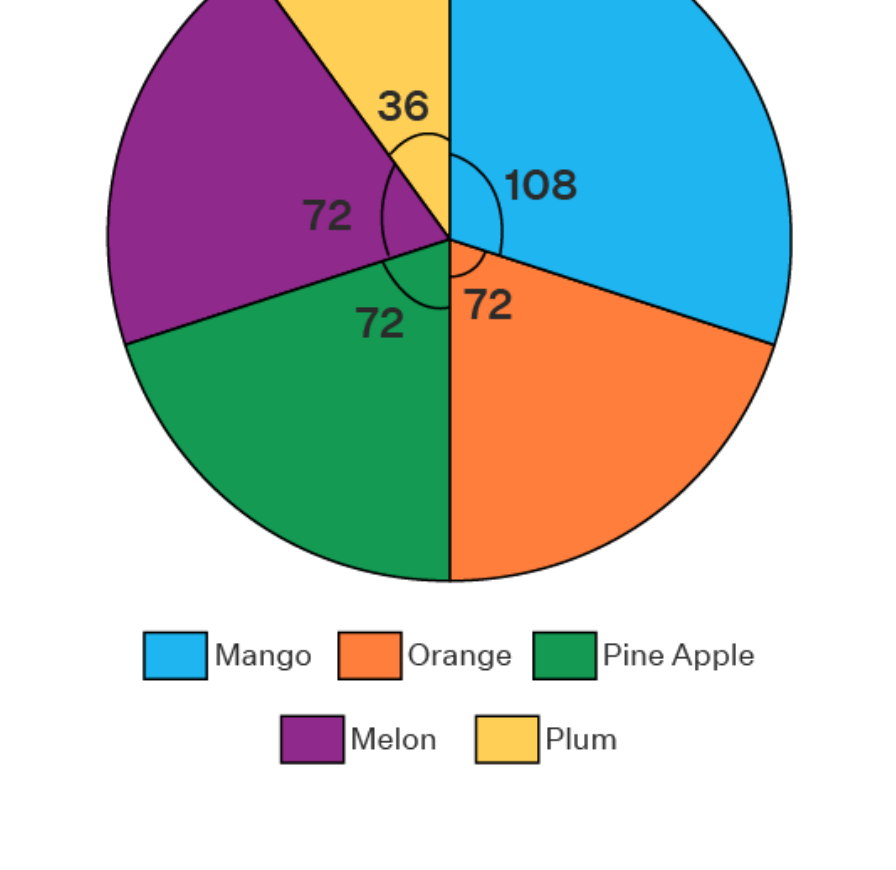
Line Graphs

In data handling the data represented in the form of a line on a graph is the [line graph](#). The graph helps in showcasing the different trends or changes in the data. The line segment plotted on the graph is constructed by connecting individual data points together. Look at the example below to understand it better.



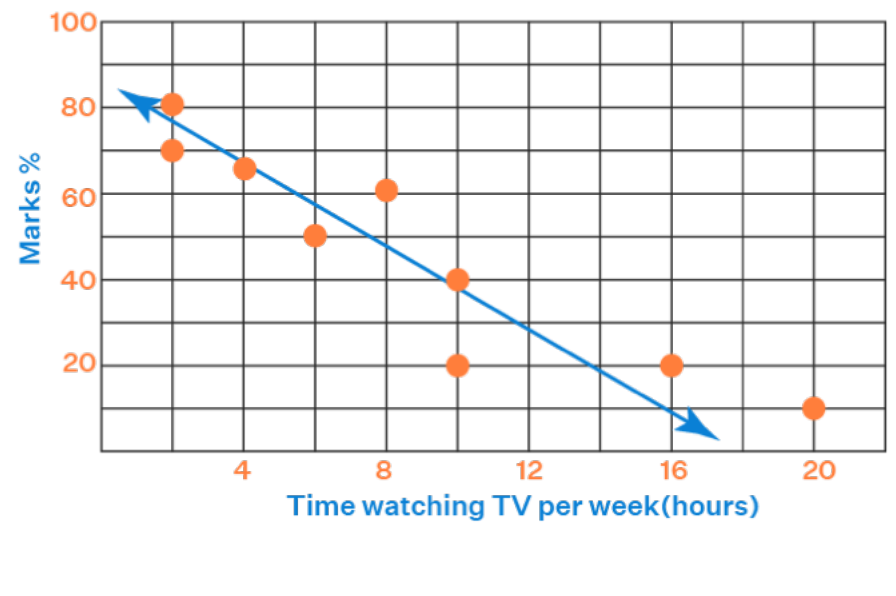
Pie Charts

A [pie chart](#) is data represented in a circular graph divided into smaller sectors to denote certain information. Pie charts help in showcasing the profit and loss for a business, while in school in showcasing the number depending on the data. This kind of chart is widely used in marketing sales. Look at the example below, the pie chart shows how people like the mentioned fruits from a group of 360.



Scatter Plot

[Scatter plot](#) represents the points and then the best fit line is drawn through some of the points. Any 3D data in data handling can be represented by a scatter plot. Look at the example below to understand it better.



Related Topic

Listed below are a few interesting topics related to data handling. Take a look.

- [Absolute Value Graph](#)
- [Frequency Distribution Table](#)
- [Probability and Statistics](#)

Examples on Data Handling

Example 1: Henry wants to introduce his 5-year-old daughter to data handling. Which type of graphical representation can he use for this?

Solution:

As his daughter is just 5 years old, he should prefer using Pictograph to introduce data handling. In this representation, simple pictures like circles, stars are drawn to represent different data.

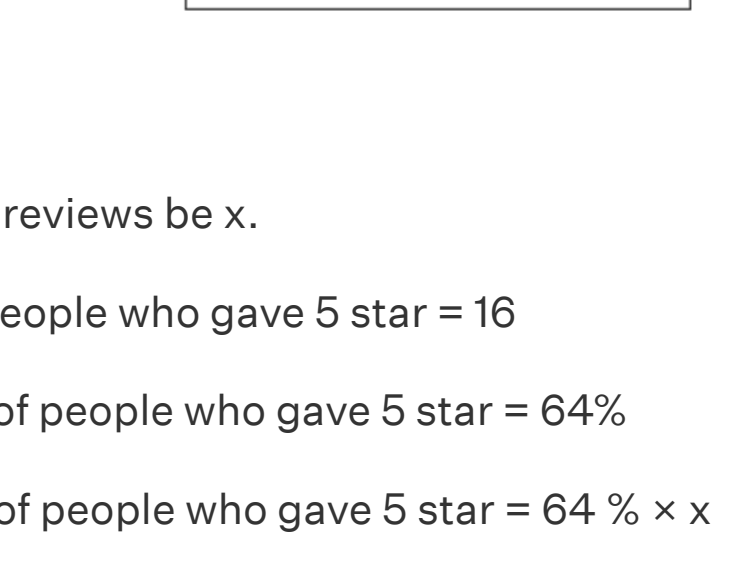
Example 2: How is data represented graphically?

Solution: Various types of graphs that can be used for representing data are:

- Bar graph
- Scatter plot
- Line graph
- Area plot
- Pie chart/ Circle chart
- Picture graph

Depending on the purpose, a suitable graph can be chosen.

Example 3: Here is a review of an electronic product. Out of all the people who gave their reviews, 16 of them gave a 5-star rating to the product. Can you find out how many people provided their feedback in all?



Solution:

Let the total reviews be x.

Number of people who gave 5 star = 16

Percentage of people who gave 5 star = 64%

So, number of people who gave 5 star = 64 % × x

16 = 64/100 × x

x = (16 × 64)/100

x = 25.

Therefore, 25 people gave reviews for the product

Practice Questions on Data Handling

FAQs on Data Handling

What is Data Handling?

Data Handling is the process of gathering, recording, and presenting information in a way that is helpful to analyze, make predictions and choices. There are two types of data handling namely quantitative data and qualitative data. Data handling can be represented through various graphs.

What are the Two Types of Data Handling?

The two types of data handling are qualitative data and quantitative data. Quantitative data gives numerical information, while qualitative data gives descriptive information about anything. Quantitative can be either discrete or continuous data.

What are the Steps Involved in Data Handling?

The six steps that are involved in data handling are:

- Purpose
- Collection of Data
- Presentation of Data
- Graphical Representation of Data
- Analyzing the Data
- Conclusion

What are the Types of Graphical Representations in Data Handling?

There are numerous types of graphical representation for the data that are available. Some of the most extensively used graphical representations are :

- Bar graph
- Scatter plot
- Line graph
- Area plot
- Pie chart/ Circle chart

What is the Difference Between Data and Information in Data Handling?

The term data refers to the collection of certain facts that are quantitive in nature like height, number of children, etc. Information on the other hand is a form of data after being processed, arranged, and presented in a form that gives meaning to the data.

What is the Difference Between the Chart and Graph?

The difference between chart and graph can be understood from the fact that - All graphs are charts but every chart is not a graph. Charts display data in the form of a diagram, table, or graph. So, the graph is just a pictorial way of presentation of information.